

Yash Rawal

+91-9399159685 | yashrawal987@gmail.com | linkedin.com/in/rawal-yash | github.com/YashRL | Ratlam (457001), Madhya Pradesh, India

Professional Summary

Applied AI & Machine Learning Engineer serving a **1M+ user scale** by architecting enterprise agentic platforms, MCP server infrastructure, and ERP AI systems across 120+ organizations. Specialized in LLM orchestration (LangGraph, ReAct loops), enterprise RAG, and Azure OpenAI deployments. Active contributor to Spring AI, bridging generative AI and model fine-tuning with production purchasing workflows.

Skills

- Languages & Backend:** Python, SQL, Java, FastAPI, Spring Boot, PostgreSQL, REST APIs
- AI & ML Engineering:** MCP Server Development, Agentic AI, RAG & Vector Search, Model Training & SFT, Context Graphs
- AI & ML Frameworks:** LangGraph, LangChain, Spring AI, PyTorch, vLLM, LlamaIndex, CrewAI, GROBID
- Platforms & Models:** Azure OpenAI, Claude, Amazon Bedrock, Google Vertex AI, Qwen, Meta Llama
- Infrastructure & Security:** Docker, Kubernetes, Linux, CI/CD, LLM-as-Judge, Red Teaming, PII Sanitization

Work Experience

Bigscal Technologies Pvt. Ltd.
AI/ML Engineer Oct 2024 - Present

- Remote
- Deployed a production-scale agentic platform and multi-agent systems serving **1M+ active ERP users** across 120+ enterprise tenants.
 - Designed a capability contract layer to execute nested profiles inside a single ReAct loop, reducing execution overhead by **3x** and token consumption by 70%.
 - Trained domain-specific **Qwen2.5-7B** models on A100 GPUs and fine-tuned E5-large-v2 embeddings using PyTorch to optimize query intent mapping.
 - Developed the Prompt Lab evaluation control plane to collect reasoning trajectories and human preference annotations, establishing a scalable **Human Feedback (RLHF) Pipeline**.
 - Implemented a hierarchical RAG pipeline indexing 10,000+ technical documents using GROBID, pdfplumber, and PostgreSQL/PGVector, achieving a **90%+ Recall@10** retrieval rate.
 - Built production guardrails against prompt injections and data leaks, integrating runtime tool validation and **automated PII redaction**.

Bigscal Technologies Pvt. Ltd.
AI/ML Engineer Intern Apr 2024 - Sep 2024

- Surat, India
- Engineered clinical summarization pipelines for 116K+ medical records with 4-bit quantized Llama 3, and designed real-time WebSocket audio streaming for Indic ASR (Whisper, Vosk, NeMo).
 - Researched audio digital signal processing (STFT, Mel-spectrograms, MFCC feature extraction via librosa/torchaudio) and CNN audio architectures (VGGish, ResNet), compiling custom Indic LJSpeech-format datasets.

Projects

- Modular Agent Harness & Memory System ("Brain")** | Open Source
Principal Creator & Architect of an open-source plug-and-play agentic framework designed as modular building blocks for flexible enterprise workflows, implementing a formal ReAct loop with step budgets, vector tool routing, and dynamic MCP settings caching.
Java, Spring Boot, PostgreSQL, WebSockets
- Spring AI (spring-projects/spring-ai)** | Open Source
Merged PR #5711 to propagate reasoningContent from AWS Bedrock and OpenAI-compatible models, and resolved Converse API prompt-caching TTL latency consistency.
Java, Spring Boot, Python
- Mini Language Model (~85M Parameter Architecture)** | Personal Research
Pretrained an ~85M-parameter transformer to ~1.43 validation loss on TinyStories; engineered FragmentStream chunked attention to eliminate $O(T^2)$ VRAM overhead on commodity GPUs and custom Triton kernels with KV caching for low-latency inference.
Python, PyTorch, CUDA, Triton, Tesla P100/T4

Education

Master of Computer Applications (Artificial Intelligence & Machine Learning) - Parul University 2022 - 2024

Certifications

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|---|-----------|
| Microsoft Certified: Azure AI Fundamentals (AI-900) | Microsoft |
| Microsoft Certified: Azure Fundamentals (AZ-900) | Microsoft |